Environmental Monitoring Data Analyzer

Python Programming Case Study

1. Problem Statement

Environmental monitoring systems collect readings such as air-quality, temperature, or pollution measurements over time. Reviewing many readings manually can be difficult. This program processes a set of numeric readings, classifies each reading as Good, Moderate, or Poor, and calculates summary statistics to help users understand the data.

2. Objective

To write a Python program that processes multiple environmental readings, calculates the average, identifies the highest and lowest values, and labels each reading according to predefined threshold ranges.

3. Assumptions

• Readings are numeric values stored in a nested list.

• Each inner list represents a group of readings.

• A reading of 50 or below is classified as Good.

• A reading above 50 and up to 75 is classified as Moderate.

• A reading above 75 is classified as Poor.

• The input contains at least one reading, so average, highest, and lowest can be calculated.

• The thresholds are illustrative and should be adjusted to match the environmental metric being monitored.

4. Algorithm

1. Start the program.

2. Initialize the nested list of environmental readings.

3. Set total and count to zero.

4. Initialize highest and lowest using the first reading.

5. Traverse each group and each reading using nested loops.

6. Add each reading to total and increment count.

7. Update highest if the current reading is greater; update lowest if it is smaller.

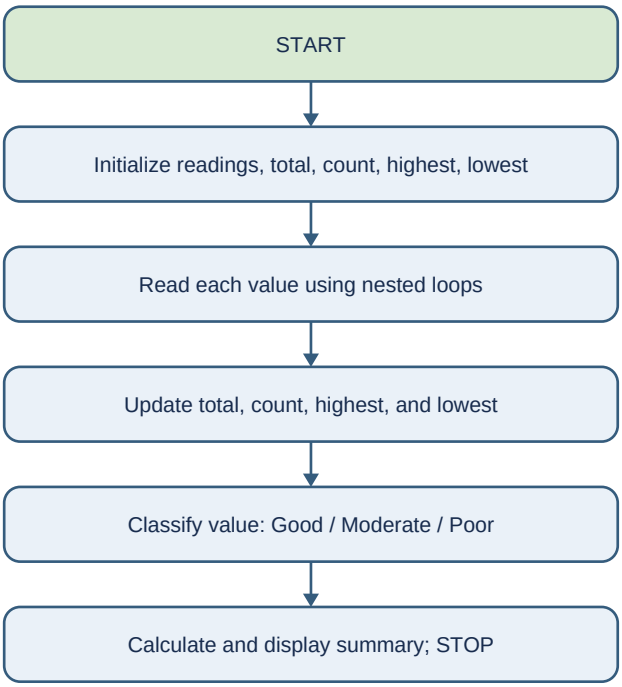
8. Classify each reading: Good if it is at most 50, Moderate if it is at most 75, otherwise Poor.

9. Calculate average as total divided by count.

10. Display the average, highest reading, and lowest reading.

11. Stop.

5. Flowchart



Repeat the reading-processing steps until all values are handled.

6. Pseudocode

```
BEGIN
    readings ← [[40, 50, 60], [70, 80, 90], [30, 45, 55]]
    total ← 0
    count ← 0
    highest ← first reading
    lowest ← first reading

    FOR each group in readings
        FOR each value in group
            total ← total + value
            count ← count + 1

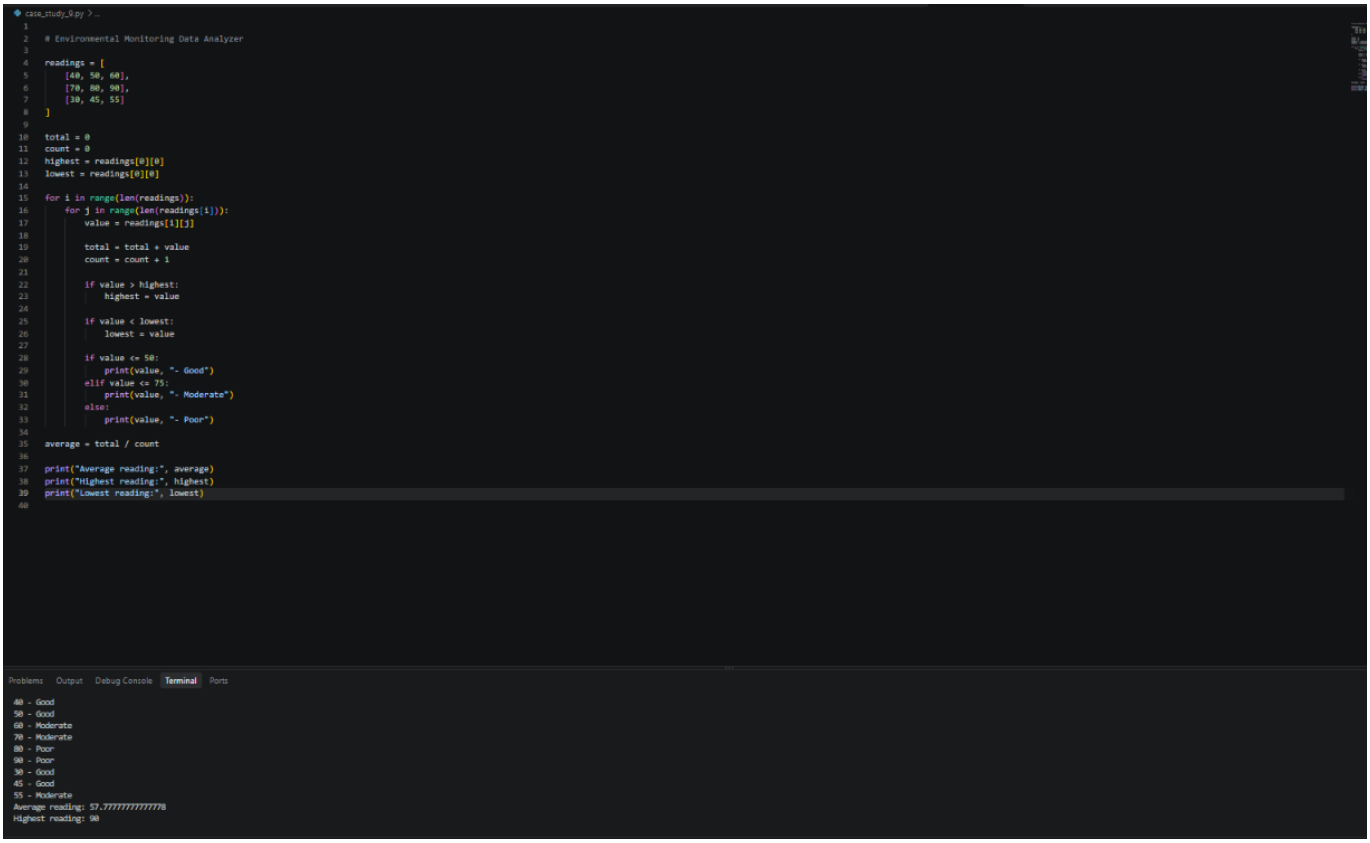
            IF value > highest
                highest ← value
            IF value < lowest
                lowest ← value

            IF value ≤ 50
                PRINT value, "Good"
            ELSE IF value ≤ 75
                PRINT value, "Moderate"
            ELSE
                PRINT value, "Poor"
        END FOR
    END FOR

    average ← total / count
    PRINT average, highest, lowest
END
```

7. Source Code and Output Screenshot

The supplied screenshot shows the Python program and its terminal output.



8. Conclusion

The Environmental Monitoring Data Analyzer demonstrates how Python can summarize and categorize environmental readings. Nested loops process every value, conditional statements assign a quality category, and simple calculations determine the average, maximum, and minimum. For the supplied data, the highest reading is 90 and the lowest is 30;

the average is about 57.78. The classification thresholds are examples and should be selected according to the specific environmental measurement and applicable standards.